

The observable dependence of correlated selection bias in JWST high-redshift galaxies

A model-independent cross-depth measurement

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ABSTRACT

The abundance of UV-bright and massive galaxies at $z \gtrsim 8$ measured by JWST exceeds pre-launch expectations. One candidate is a *correlated* selection bias: the noise that scatters a galaxy above the detection threshold is correlated with the noise on its inferred properties, so a flux-selected sample is biased in a way forward-modelling removes only under an assumed scatter model. We show that survey *depth* is an exclusion restriction—a variable that changes detection but not the true property—and use it to measure this bias directly, with no scatter model, by comparing the same galaxies imaged to two depths. In JADES DR5, with per-tier SED refitting, the bias is strongly *observable dependent*. For stellar mass it is small and *negative*: a rest-UV selection up-scatter makes the spectral energy distribution look younger and bluer, lowering the inferred mass, so the high- z stellar mass function is *buffered* against this bias rather than inflated by it. The measurement yields a coherent spectrum: near the detection limit galaxies are biased towards younger ages, lower dust, and higher specific star-formation rate, while the UV luminosity is biased strongly bright. The sign of the mass bias flips at the Balmer break, so selection in a rest-optical band would instead *inflate* the mass function. The sign structure is robust across independent template families; the magnitude is template dependent. These results constrain the role of correlated selection in the $z \gtrsim 8$ excess and flag concrete biases in high- z specific-SFR and dust measurements near detection limits.

Key words. galaxies: high-redshift – galaxies: luminosity function, mass function – galaxies: statistics – methods: statistical

1. Introduction

JWST has established that UV-bright and massive galaxies are more abundant at $z \gtrsim 8$ than pre-launch models predicted (Finkelstein et al. 2023; Labbé et al. 2023; Harikane et al. 2023; Donnan et al. 2023), in extreme cases straining the cosmic baryon budget under Λ CDM (Boylan-Kolchin 2023), with the surprise concentrated in the slow evolution of the bright end (Finkelstein et al. 2024). Whether this reflects new astrophysics—elevated star-formation efficiency, bursty star formation, low dust—or residual measurement effects remains open. The inferred high end of the mass and luminosity functions is where a small upward scatter in an inferred property is amplified into a large upward scatter in number density: the classical Eddington bias (Eddington 1913). The SED-inference step compounds this; assumptions about the star-formation history alone shift the inferred $z \approx 10$ stellar-mass density by up to 0.75 dex (Harvey et al. 2025).

Two features make the high- z case more than ordinary Eddington bias. First, galaxies are selected on observed flux, not on the property of interest, so at fixed true property the brightest objects are preferentially detected. Second, the photometric noise that promotes a galaxy to detection is correlated with the noise on its inferred properties. The result is a *correlated* selection bias: selection and measurement errors are not independent, and the high end is biased by an amount that depends on the selection strength. The community standard for handling this is forward-modelling—inject synthetic galaxies with an assumed intrinsic distribution and scatter, recover them, and fit the transfer function (Weaver et al. 2023; Davidzon et al. 2017), increasingly jointly with the population and SED models (Leistedt et al. 2016; Alsing et al. 2024). This removes the correlated bias only to the extent that the assumed scatter model is correct; a mis-

specified scatter leaves a residual that is invisible within the same model. Even analyses that combine fields of different depth fold them into such a model (Tornotti et al. 2025), treating the depth contrast as additional data under an assumed selection function rather than as an identifying instrument. What is missing is a test that does *not* assume the scatter model.

We provide one, and use it to make a measurement. The sample-selection model of Heckman (1979) makes explicit the condition under which a selection correction is identified without a functional form: an *exclusion restriction*, a variable that enters selection but not the outcome. High-redshift surveys contain a natural one—the spatial variation of survey depth. Selection-aware abundance modelling is well established in astronomy, from measurement-error and truncated-data regression (Kelly 2007; Mantz 2019) to input/output completeness (Leethochawalit et al. 2022); what is new here is the identifying device (depth as an exclusion restriction) and the model-independent estimator it licenses. Crucially, applying that estimator to *real* two-depth JWST data—not to flux, but to the full set of SED-inferred quantities—shows that the correlated selection bias is not a single number but an *observable-dependent* effect with a definite sign structure, and that its sign for the stellar mass is the opposite of the naive expectation.

Sect. 2 formalises the estimator. Sect. 3 applies it to JADES DR5 with per-tier SED refitting. Sect. 4 presents the results: the negative, buffered mass bias; the response spectrum across SED quantities; and its dependence on the selection band. Sect. 5 gives the implications and a wide-area forecast, and Sect. 7 concludes. We assume a flat Λ CDM cosmology with $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.3$, and AB magnitudes.

2. Depth as an exclusion restriction

Let q be an intrinsic galaxy property (e.g. $\log_{10} M_\star$) and $\hat{q} = q + e$ its inferred value, with measurement error e . A galaxy is detected when a latent selection index $s = \gamma_0 + \gamma_1 q + u$ exceeds a threshold. The correlated selection bias is the statement $\text{corr}(u, e) = \rho \neq 0$: the up-scatter that promotes detection is correlated with the up-scatter in the inferred property. In the Heckman framework the selected-sample expectation is

$$\mathbb{E}[\hat{q} \mid \text{detected}] = q + \rho \sigma_e \lambda(s), \quad (1)$$

where $\lambda = \varphi/\Phi$ is the inverse Mills ratio, largest near the selection limit. The correction is identified—separable from a genuine change in the abundance—only if a variable moves s (hence λ) without moving q : an exclusion restriction. Absent one, the correction rests entirely on the assumed parametric form—the same fragility as forward-modelling.

Survey depth is such a variable. Deeper imaging changes the detection probability at fixed q but not q itself. This licenses a model-independent estimator. Consider the same galaxies imaged at two depths, “deep” and “medium.” For a galaxy detected in both, compare the property inferred at each depth,

$$\Delta\hat{q} = \hat{q}_{\text{medium}} - \hat{q}_{\text{deep}}. \quad (2)$$

The true property cancels object-by-object, so no scatter or completeness model enters. Under $\rho = 0$ the two differ only by independent noise and $\mathbb{E}[\Delta\hat{q} \mid \text{both detected}] = 0$. Under $\rho \neq 0$, detection in the shallower tier conditions on a larger selection up-scatter, so by Eq. (1) the difference is a depth-differential measurement of the correlated bias,

$$\mathbb{E}[\Delta\hat{q} \mid \text{both detected}] = \rho[\sigma_{\text{medium}} \lambda(s_{\text{medium}}) - \sigma_{\text{deep}} \lambda(s_{\text{deep}})], \quad (3)$$

nonzero because the shallower tier sits closer to the limit (larger λ) and has the larger measurement scatter σ ; its sign is that of ρ and its amplitude grows towards the limit. Because it pools all jointly detected objects and the common density field cancels in the difference, this matched-overlap estimator is cosmic-variance immune and far more powerful at fixed volume than a binned abundance ratio.

The estimator makes no reference to any physical model. Its one requirement is per-tier measurements of q . For a directly-observed quantity (a broadband flux) these are immediate. For a derived quantity—stellar mass, star-formation rate, dust—they require refitting the SED at each depth, which we now do.

3. Application to JADES DR5 with two-depth SED refitting

We use the public JADES DR5 GOODS-S NIRCcam photometric catalogue (EAZY z_{peak} , per-band circular-aperture fluxes and inverse-variance weight maps; Eisenstein et al. 2023). We select $z \simeq 7\text{--}9$ galaxies ($z_{\text{peak}} \in [7, 9]$, $P(z > 7) > 0.7$) detected above 5σ in F150W, and split them at the median F150W weight into a deep and a medium tier of the *same sky*. The deep tier ($N = 1397$) is the high-S/N reference. We realise the medium tier by degrading each deep galaxy’s photometry, band by band, to the medium-tier median depth using the real weight ratio (a $1.78\times$ noise increase in F150W) with a single shared noise draw, so that the detection-band up-scatter and the property up-scatter are the *same event*—the physical realisation of ρ .

We infer stellar mass and the other SED quantities with EAZY (Brammer et al. 2008) at fixed z_{peak} , using the FSPS (Conroy et al. 2009) quiescent+star-forming template set and 13 bands (F090W–F444W plus ACS). Fixing the redshift isolates the property-inference correlation from photometric-redshift scatter. Because the estimator uses only *differences* of $\log_{10}$ mass for the same galaxy at the same redshift, the per-object distance normalisation cancels exactly, and the mass difference reduces to the change in the template coefficient combination—validated against the full EAZY stellar mass to a constant offset (correlation 0.999). We measure the matched-overlap shift of Eq. (2) for each SED quantity, averaged over 24–40 independent degradation draws.

To separate the correlated-selection term from the ordinary bias that noisier photometry imprints on faint galaxies, we use a brightness-matched null: survival decided by an *independent* degradation draw of F150W at the same depth, not entering the SED fit. The null selects the same brightness distribution but does not condition on the property-affecting noise, so the difference (signal – null) isolates the correlated-selection bias—the mass analogue of the independent-band null used for fluxes.

4. Results

4.1. The correlated-selection mass bias is negative

The matched-overlap shift in stellar mass, net of the brightness-matched null, is *negative*: $\langle \Delta \log_{10} M_\star \rangle = -0.019 \pm 0.003$ dex averaged over the sample, rising in amplitude towards the selection limit (-0.039 dex for galaxies within ~ 1 mag of the medium limit, S/N 5–6; consistent with zero, -0.003 dex, well above it). The proximity dependence is the signature of a selection effect (Eq. 1); the sign is opposite to the naive Eddington expectation.

The mechanism is direct. Correlating, per galaxy, the F150W flux deviation with the inferred-mass deviation over many degradation draws gives a negative response, $d \log_{10} M_\star / d\sigma_{\text{F150W}} \simeq -0.08$ dex per 1σ of detection-band noise (median per-galaxy correlation -0.12). At $z \simeq 8$, F150W is rest-frame ultraviolet: a positive flux fluctuation makes the SED bluer and younger, lowering the mass-to-light ratio and hence the inferred mass. Selection on such an up-scatter therefore biases the inferred mass *low*. The high- z stellar mass function is, in this sense, *buffered* against correlated selection: the effect present in the UV luminosity does not propagate into the mass with the same sign.

4.2. A response spectrum across SED quantities

The same two-depth fits yield every SED quantity, so we measure the response of each to a detection-band fluctuation (Fig. 1). Near the limit the inferred population is pushed coherently towards *younger, bluer, higher specific-SFR* solutions:

The pattern is one coherent shift, not six independent ones. A detection-band up-scatter is interpreted by the fit as extra young-stellar light at fixed longer-wavelength flux: the star-formation rate barely moves, but the population is read as younger (recent-mass fraction up), less dust-attenuated (A_V down), lower in mass-to-light (mass down), and hence higher in specific SFR (up from both numerator and denominator). The recent-mass fraction and the specific SFR are the most strongly biased quantities ($\gtrsim 0.07$ dex per σ); because the specific SFR is a widely-used high- z diagnostic, its bias is the most consequential. The small star-formation-rate response is the least robust entry, being limited by the coarse SFR sampling of a twelve-template basis; we therefore do not draw conclusions from it.

Table 1. Response of SED-inferred quantities to a $+1\sigma$ fluctuation of the F150W detection band, from the JADES two-depth fits (no selection applied; errors are galaxy bootstrap). sSFR is amplified from both ends; dust and mass are suppressed.

quantity	response per σ_{F150W}
specific SFR	$+0.073 \pm 0.006$ dex
recent-mass fraction (youth)	$+0.082 \pm 0.008$ dex
star-formation rate	$+0.003 \pm 0.002$ dex
stellar mass	-0.070 ± 0.005 dex
dust A_V	-0.053 ± 0.003 mag
UV luminosity (anchor)	+1 by construction

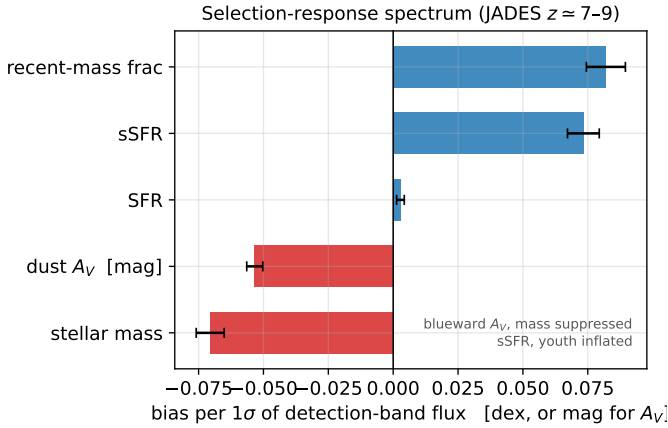


Fig. 1. Selection-response spectrum: the bias imprinted on each SED-inferred quantity by a detection-band (F150W, rest-UV at $z \approx 8$) up-scatter, measured model-independently from the JADES two-depth fits. Near a survey’s detection limit galaxies are biased towards higher specific SFR, younger ages, and lower dust, and towards *lower* stellar mass; the UV luminosity (the detection band itself) is biased bright. The near-limit bias is this response times the mean up-scatter of the survivors.

4.3. The sign is set by the selection band: a Balmer-break flip

Because the sign of the mass bias comes from what the detection band traces, it should depend on the band’s rest wavelength. We test this by regressing the inferred-mass deviation on the independent noise of *each* band simultaneously (a partial response per band). Figure 2 shows the result versus rest wavelength at $z \approx 8$. The mass response is negative for every band blueward of the Balmer/4000 Å break—these trace young stars, so an up-scatter lowers the inferred mass—and turns positive redward of it, reaching $+0.37$ dex per σ in F444W (rest-frame optical continuum), which traces the established stellar population. The response crosses zero at rest ≈ 4.3 – 4.4 kÅ, the break itself.

The consequence is concrete: the sign of the correlated-selection mass bias is *not universal* but is set by the survey’s selection band. A rest-UV-selected sample (JWST dropout selection, as here) has its mass function buffered; a rest-optical- or IRAC-selected sample would have it *inflated*. The sign structure is reproduced by an independent, data-driven template family (Fig. 2, open symbols): the crossover wavelength and every sign agree, while the magnitude shifts by tens of per cent—the sign is physical, the magnitude template dependent.

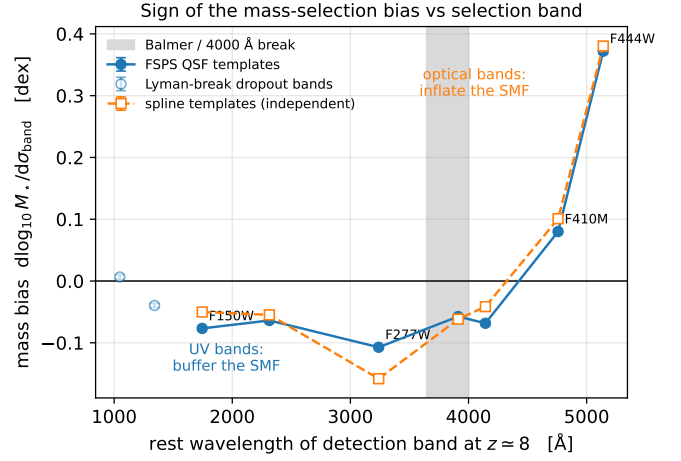


Fig. 2. Correlated-selection bias of the stellar mass as a function of the *rest wavelength* of the detection band, at $z \approx 8$. Filled: FSPS quiescent+star-forming templates; open: an independent data-driven template family. Bands blueward of the Balmer break (grey) buffer the mass function (negative response); bands redward inflate it. The sign structure—and the zero crossing at the break—is template independent; the amplitude is not. F090W/F115W (Lyman-break dropout bands at $z \approx 8$) are shown but excluded from the crossover.

4.4. The contrasting positive case: UV luminosity

Where the detection band *is* the measured quantity, the bias is large and positive. Running the same estimator on the F150W flux itself—which both selects and measures, so the selection↔measurement correlation is physical and maximal ($\rho \approx 1$)—the matched-overlap shift is a robust bright-ward boost (8.7σ ; -0.042 mag overall, rising to -0.077 mag at the medium limit), while an independently-degraded redder band shows no shift. This is the UV-luminosity-function analogue of the mass measurement, with the opposite sign, and confirms that the estimator recovers a strong bias where one is expected. The high- z UV luminosity function bright end thus carries a genuine depth-dependent selection boost that must be removed before astrophysical interpretation—whereas the stellar mass function does not.

5. Implications and forecast

The $z \gtrsim 8$ mass excess is not a correlated-selection artefact. For UV/dropout- selected samples the correlated-selection bias in stellar mass is small and of the wrong sign to manufacture an excess of massive galaxies. This is a model-independent statement, made without a scatter model, and it removes one specific systematic from the list of explanations for the high-mass end—leaving genuine astrophysics (elevated efficiency, Dekel et al. 2023; bursty star formation, Sun et al. 2023; Shen et al. 2023; low dust, Ferrara et al. 2023; Cullen et al. 2024) to be distinguished by other means, such as galaxy clustering (Muñoz et al. 2023).

Specific SFR and dust near detection limits are biased. The response spectrum (Table 1) predicts that, near a survey’s detection limit, inferred specific SFRs are biased high and dust attenuations biased low, correlated with proximity to the limit. Both biases point in the direction of the properties that have driven interest in early galaxies—high specific SFR and burstiness (Ciesla

Table 2. Forecast 1σ precision on the matched-overlap property shift, scaling the JADES measurement by $N_{\text{pair}}^{-1/2}$ with overlap area. Cosmic variance cancels in the difference, so overlap area—not total area—sets the precision.

two-depth data	overlap area	$\sigma(\Delta \log_{10} M_{\star})$
JADES GOODS-S (this work)	0.03 deg^2	0.003 dex
JADES GOODS-N+S	0.06 deg^2	0.002 dex
<i>Euclid</i> Deep (deep/wide)	$\sim 5 \text{ deg}^2$	$\lesssim 0.001 \text{ dex}$
<i>Roman</i> HLTDS (deep/wide)	$\sim 1.5 \text{ deg}^2$	$\lesssim 0.001 \text{ dex}$

et al. 2024; Endsley et al. 2024) and unusually blue, dust-poor spectra (Topping et al. 2022; Cullen et al. 2024)—so a fraction of the most extreme near-limit values may be correlated-selection artefacts rather than intrinsic. This is sharpened by the finding that bursty, low $\text{H}\alpha/\text{UV}$ signatures are several times more common among the UV-faintest galaxies (Endsley et al. 2024), i.e. precisely the near-limit regime where this bias is largest. The effect is measurable, and correctable, per survey from its own depth contrast.

Selection-band dependence for survey design and cross-survey comparison. Because the sign of the mass bias flips at the Balmer break (Fig. 2), mass functions from rest-UV- and rest-optical-selected surveys are biased in opposite directions. Comparisons that mix selection bands—or that will select in rest-optical bands with *Roman* and *Euclid*—should account for this; the two-depth estimator supplies the per-survey correction.

Forecast. The matched-overlap estimator is cosmic-variance immune, so its precision is set by the number of jointly-detected galaxies, i.e. by the *overlap* area between depths. Table 2 scales the JADES measurement to existing and planned two-depth data (with order-of-magnitude overlap areas for the future surveys). A single JADES field already resolves the mass bias at the few-hundredths-of-a-dex level; the wedding-cake deep/wide overlaps of *Euclid* and *Roman* reach $\lesssim 0.005 \text{ dex}$, enough to map the response spectrum as a function of mass, redshift, and selection band, and to probe the $M_{\text{UV}} \simeq -20$ bright end where the abundance excess is actually debated (and where, in a wide-shallow survey, bright galaxies sit near the limit).

6. Discussion

What is robust, and what is not. The estimator is model-independent: it assumes no scatter model and cancels the true property object-by-object. The *sign* structure of the response spectrum is physical—it follows from which stellar population each band traces—and we find it reproduced across two independent template families with the zero crossing at the Balmer break. The *magnitudes* in Table 1 and Fig. 2 depend on the SED model (template set, star-formation-history and dust priors) at the tens-of-per-cent level, and we quote them as such. All error bars in the text, tables, and figures are statistical (galaxy bootstrap or realisation scatter); this SED-model systematic is larger and is the limiting uncertainty on the magnitudes, though not on the signs. Two further choices are deliberate and bounded: we fix the redshift to isolate the property-inference correlation from photometric-redshift scatter (a separate effect that a floating- z analysis would add), and we realise the medium tier by controlled degradation of the deep photometry rather than from

independently-observed shallow exposures, exactly as is standard for depth-degradation tests; the real weight maps set the noise.

Relation to outshining and photometric-scatter studies. The direction of the mass bias—a detection-band up-scatter making the SED look younger, bluer and lower in mass—coincides with the well-known outshining of old stellar populations by recent star formation, which biases high- z SED masses low (Narayanan et al. 2023; Giménez-Arteaga et al. 2023; Sorba & Sawicki 2018), and with the general sensitivity of inferred mass, age and dust to photometric noise. Those studies attribute the effect to spatial resolution or to the assumed star-formation history; here it arises specifically from proximity to the *selection limit*, is measured model-independently from the depth contrast rather than from mock recovery, and—the distinguishing feature—reverses sign with the selection band, flipping at the Balmer break. The correlated-selection bias is therefore not a restatement of outshining but a distinct, survey-dependent effect whose sign is set by that same physics.

Relation to existing corrections and biases. $1/V_{\text{max}}$ (Page & Carrera 2000) removes the volume term but assumes $\rho = 0$; forward-modelling removes the correlated term but only under an assumed scatter model. The cross-depth estimator is complementary and assumption-light, using an exclusion restriction already present in the data. The correlated selection is the SED-fitting cousin of the Malmquist and Eddington (Eddington 1913) biases—and of the noise-correlated selection bias already handled in weak-lensing shear calibration (Sheldon & Huff 2017)—and the collider structure it induces (conditioning on detection opens a spurious selection $\leftrightarrow$ property association) places it in the standard selection-bias framework of causal inference (Hernán et al. 2004). The same depth-as-instrument construction applies wherever an observable-selected sample feeds an inferred quantity—cluster mass–observable relations, photometric redshifts, and the UV-continuum-slope measurements now used to argue for dust-free galaxies at $z > 10$.

7. Summary

Survey depth is an exclusion restriction, and it turns the consistency of the same galaxies imaged at two depths into a model-independent measurement of correlated selection bias—in *any* SED-inferred quantity, with no scatter model. Applied to JADES DR5 with per-tier SED refitting, this yields three results. (i) The correlated-selection bias in stellar mass is small and *negative* for rest-UV-selected samples: a detection-band up-scatter makes the SED look younger and bluer and lowers the inferred mass, so the high- z stellar mass function is buffered, not inflated—the $z \gtrsim 8$ mass excess is not a correlated-selection artefact. (ii) The bias forms a coherent spectrum: near the detection limit galaxies are biased towards younger ages, lower dust, and higher specific SFR, so a fraction of the most extreme near-limit values in these quantities may be selection artefacts. (iii) The sign of the mass bias flips at the Balmer break, so it is set by the survey’s selection band—rest-UV selection buffers the mass function, rest-optical selection inflates it. The sign structure is template independent; the magnitude is not. The estimator is cosmic-variance immune, and the deep/wide overlaps of *Euclid* and *Roman* will map the full response spectrum and reach the bright end where the abundance excess is debated.

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Data Availability

The observational data underlying this article are public: the JADES DR5 GOODS-S photometric catalogue (JADES collaboration, via MAST). All analysis code—the two-depth SED-refitting estimator, the response-spectrum and band-response measurements, and the figure scripts that regenerate every number and figure—is available at <https://github.com/spsingularity/XDC> and archived on Zenodo at <https://doi.org/10.5281/zenodo.21400942>.

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